



习题 1.19

$$I_1 = \int_0^{\infty} f(x) \cos mx \, dx = \frac{1}{2} \int_{-\infty}^{+\infty} f(x) \cos mx \, dx = \frac{1}{2} \int_{-\infty}^{+\infty} f(x) (e^{imx} + e^{-imx}) \cdot \frac{1}{2} \, dx$$

$$= \frac{1}{2} \int_{-\infty}^{+\infty} f(x) e^{imx} \, dx = \pi i \operatorname{Res} [f(x) e^{imx}] = \pi i \sum_{j=0}^{+\infty} \frac{a_{-1-j}}{j!} (im)^j$$

$$\left(\begin{aligned} \text{设 } f(x) &= \sum_{k=-\infty}^{+\infty} a_k x^k, \quad e^{imx} = \sum_{k=0}^{+\infty} \frac{(imx)^k}{k!} \\ f(x) e^{imx} &= \sum_{k=-\infty}^{+\infty} \sum_{j=0}^{+\infty} \frac{a_k}{j!} (im)^j x^{k+j}, \quad \operatorname{Res} [f(x) e^{imx}] = \sum_{j=0}^{+\infty} \frac{a_{-1-j}}{j!} (im)^j \end{aligned} \right)$$

$$I_2 = \int_0^{\infty} G(x) \sin mx \, dx = \frac{1}{2} \int_{-\infty}^{+\infty} G(x) \sin mx \, dx = \frac{1}{2} \int_{-\infty}^{+\infty} G(x) \frac{e^{imx} - e^{-imx}}{2i} \, dx$$

$$= \frac{1}{2i} \int_{-\infty}^{+\infty} G(x) e^{imx} \, dx = \pi \operatorname{Res} [G(x) e^{imx}] = \pi \sum_{j=0}^{+\infty} \frac{b_{-1-j}}{j!} (im)^j$$

$$\text{设 } G(x) = \sum_{k=-\infty}^{+\infty} b_k x^k$$

习题 1.20

$$(1) \int_0^{2\pi} \frac{dx}{(1+\varepsilon \cos x)^2}, \quad 0 < \varepsilon < 1$$

$$\text{解: 原积分} = \int_{|z|=1} \frac{\frac{1}{iz} dz}{(1+\varepsilon(z+z^{-1}))^2} = \frac{1}{i} \int_{|z|=1} \frac{dz}{z(1+\varepsilon(z+z^{-1}))^2} = \frac{1}{i} \int_{|z|=1} \frac{dz}{z(1+\varepsilon(z^2+z^{-1})+2\varepsilon(z+z^{-1}))}$$

$$= \frac{1}{i} \int_{|z|=1} \frac{dz}{\varepsilon z^3 + 2\varepsilon z^2 + 2}$$

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习题 1.20

$$(1) \int_0^{2\pi} \frac{dx}{(1+\varepsilon \cos x)^2} = \int_{|z|=1} \frac{dz}{\lambda z \left[1+\varepsilon \frac{(z+z^{-1})}{2}\right]^2} = \frac{1}{\lambda} \int_{|z|=1} \frac{dz}{\left[\frac{\varepsilon z^2}{2} + z + \frac{-\varepsilon z^{-1}}{2}\right]^2} = \frac{4}{\lambda} \int_{|z|=1} \frac{z dz}{(\varepsilon z^2 + 2z + \varepsilon)^2}$$

奇点 $z_0 = \frac{-2 \pm \sqrt{4-4\varepsilon^2}}{2\varepsilon} = \frac{-1 \pm \sqrt{1-\varepsilon^2}}{\varepsilon} = -\frac{1}{\varepsilon} \pm \sqrt{\left(\frac{1}{\varepsilon}\right)^2 - 1}$

要满足 $|z_0| < 1$, 则 $z_0 = \frac{\sqrt{1-\varepsilon^2}-1}{\varepsilon} = -\frac{\varepsilon}{(\sqrt{1-\varepsilon^2}+1)}$ (= 留点). 记 $z_1 = \frac{-1-\sqrt{1-\varepsilon^2}}{\varepsilon}$

$$f(z) = \frac{z}{(\varepsilon z^2 + 2z + \varepsilon)^2}$$

$$\text{Res} f(z_0) = \frac{d}{dz} \left(\frac{z}{\varepsilon^2 (z-z_1)^2} \right) \Big|_{z_0} = \frac{1}{\varepsilon^2} \cdot \frac{(z-z_1)^2 - 2z(z-z_1)}{(z-z_1)^4} \Big|_{z_0} = \frac{1}{\varepsilon^2} \cdot \frac{z_0 - z_1 - 2z_0}{(z_0 - z_1)^2}$$

$$= \frac{1}{\varepsilon^2} \cdot \left(-\frac{z_0 + z_1}{(z_0 + z_1)^2 - 4z_0 z_1} \right) = \frac{1}{\varepsilon^2} \cdot \frac{z_0 + z_1}{4z_0 z_1 - (z_0 + z_1)^2} = \frac{1}{\varepsilon^2} \cdot \frac{-\frac{2}{\varepsilon}}{4 - \frac{4}{\varepsilon^2}} = \frac{\frac{1}{\varepsilon}}{2 - 2\varepsilon^2} = \frac{1}{2\varepsilon(1-\varepsilon^2)}$$

$$\Rightarrow \int_0^{2\pi} \frac{dx}{(1+\varepsilon \cos x)^2} = \frac{4}{\lambda} \cdot 2\pi i \cdot \text{Res} f(z_1) = 8\pi \cdot \frac{1}{2\varepsilon(1-\varepsilon^2)} = \frac{4\pi}{\varepsilon(1-\varepsilon^2)}$$

$$(2) \int_0^{2\pi} \frac{\sin^2 x}{a+b \cos x} dx = \int_0^{2\pi} \frac{\frac{1}{2}(1-\cos 2x)}{a+b \cos x} dx = \frac{1}{2} \int_0^{2\pi} \frac{1}{a+b \cos x} dx - \frac{1}{2} \int_0^{2\pi} \frac{\cos 2x}{a+b \cos x} dx$$

$$= \frac{1}{2a} \int_0^{2\pi} \frac{1}{1+\frac{b}{a} \cos x} dx - \frac{1}{2} \int_{|z|=1} \frac{\frac{1}{2}(z^2+z^{-2}) \cdot \frac{dz}{\lambda z}}{a+b \frac{(z+z^{-1})}{2}} = \frac{1}{2a} \cdot \frac{2\pi}{\sqrt{1-(\frac{b}{a})^2}} - \frac{1}{2a} \int_{|z|=1} \frac{\frac{1}{\lambda} (z^2+z^{-2}) dz}{\varepsilon z^2 + 2z + \varepsilon}, \varepsilon = \frac{b}{a}$$

$$= \frac{\pi}{\sqrt{a^2-b^2}} + \frac{\lambda}{2a} \int_{|z|=1} \frac{z^2+z^{-2}}{\varepsilon z^2 + 2z + \varepsilon} dz = \frac{\pi}{\sqrt{a^2-b^2}} + \frac{\lambda}{2a} \int_{|z|=1} \frac{z^4+1}{z^2(\varepsilon z^2+2z+\varepsilon)} dz$$

$$g(z) = \frac{z^4+1}{z^2(\varepsilon z^2+2z+\varepsilon)}, \text{ 奇点 } z_0 = \frac{-1+\sqrt{1-\varepsilon^2}}{\varepsilon}, 0, \frac{-1-\sqrt{1-\varepsilon^2}}{\varepsilon}, z_2$$

$$\text{Res} g(z_0) = \frac{z_0^4+1}{z_0^2 \left(\frac{2\sqrt{1-\varepsilon^2}}{\varepsilon} \right)} = \frac{z_0^2+z_0^{-2}}{\left(\frac{2\sqrt{1-\varepsilon^2}}{\varepsilon} \right)} = \frac{\left(\frac{2}{\varepsilon} \right)^2 - 2}{\frac{2\sqrt{1-\varepsilon^2}}{\varepsilon}} = \frac{2\left(\frac{1}{\varepsilon} \right)^2 - 1}{\sqrt{1-\varepsilon^2}} \cdot \varepsilon = \frac{2\varepsilon^{-1}-\varepsilon}{\sqrt{1-\varepsilon^2}}$$



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$$\operatorname{Res} g(z_2) = \frac{d}{dz} \frac{z^4+1}{\varepsilon z^4+2z+\varepsilon} \Big|_{z_2} = \frac{4z^3(\varepsilon z^2+2z+\varepsilon) - (z^4+1)(2\varepsilon z+2)}{(\varepsilon z^4+2z+\varepsilon)^2} \Big|_{z_2} = -\frac{2}{\varepsilon^2}$$

$$\Rightarrow \int_0^{2\pi} \frac{\sin^2 x dx}{a+b \cos x} = \frac{\pi}{\sqrt{a^2-b^2}} + \frac{1}{2a} \cdot 2\pi i \left(\frac{2\varepsilon^{-1}-\varepsilon}{\sqrt{1-\varepsilon^2}} - 2\varepsilon^2 \right) = \frac{\pi}{\sqrt{a^2-b^2}} - \frac{\pi}{a} \left(\frac{2(\frac{a}{b}) - \frac{b}{a}}{\sqrt{1-(\frac{b}{a})^2}} - 2(\frac{a}{b})^2 \right)$$

$$= \frac{\pi}{\sqrt{a^2-b^2}} \left(1 - 2(\frac{a}{b}) + \frac{b}{a} \right) + \frac{2\pi}{a} \left(\frac{a}{b} \right)^2$$

$$(3) \int_0^{2\pi} \frac{\cos x dx}{1-2\varepsilon \cos x + \varepsilon^2} = \int_{|z|=1} \frac{\frac{1}{2\lambda}(1+z^{-2}) dz}{1-\varepsilon(z+z^{-1})+\varepsilon^2} = \int_{|z|=1} \frac{\frac{1}{2\lambda}(z+z^{-1}) dz}{\varepsilon^2 z - \varepsilon z^2 - \varepsilon + z}$$

$$= \int_{|z|=1} \frac{\frac{1}{2\lambda}(z+z^{-1}) dz}{(\varepsilon z-1)(\varepsilon-z)} = \int_{|z|=1} \frac{1}{2\lambda} \cdot \frac{z^2+1}{z(\varepsilon z-1)(\varepsilon-z)} dz = \pi \left(-\frac{1}{\varepsilon} + \frac{(\frac{1}{\varepsilon})^2+1}{\frac{1}{\varepsilon}(\varepsilon-\frac{1}{\varepsilon})} + \frac{\varepsilon^2+1}{\varepsilon(\varepsilon-1)} \right)$$

$$= \pi \left(-\frac{1}{\varepsilon} + \frac{\varepsilon^2+1}{\varepsilon^2-1} + \frac{\varepsilon^2+1}{\varepsilon(\varepsilon-1)} \right) = \pi \left(-\frac{1}{\varepsilon} + \frac{\varepsilon^2+1}{\varepsilon(\varepsilon-1)} \right) = \frac{\pi}{\varepsilon} \left(\frac{\varepsilon^2-\varepsilon+2}{\varepsilon(\varepsilon-1)} \right)$$

$$(4) \int_0^{2\pi} \cos^{2n} x dx = \int_{|z|=1} \left(\frac{z^{2n}+z^{-2n}}{2} \right) \cdot \frac{dz}{\lambda z} = \frac{1}{2\lambda} \int_{|z|=1} (z^{2n-1} + z^{-2n-1}) dz$$

~~当 $n=0$ 时, 积分 $= 2\pi$~~

~~当 $n \rightarrow \infty$ 时, 积分 $= \frac{1}{2\lambda} \int_{|z|=1} z^{-2n-1} dz = \pi \frac{d}{dz} z^{-2n} \Big|_1 = 0$~~

$$(4) \int_0^{2\pi} \cos^{2n} x dx = \int_{|z|=1} \left(\frac{z+z^{-1}}{2} \right)^n \cdot \frac{dz}{\lambda z}$$

当 $n=0$ 时, 积分 $= 2\pi$; 当 n 是奇数时, $\left(\frac{z+z^{-1}}{2} \right)^n \cdot \frac{1}{2}$ 没有 z^{-1} 项 $\Rightarrow$ 积分 $= 0$

~~当 $n \rightarrow \infty$ 时, 积分 $=$~~

$$\text{当 } n \text{ 是偶数时, 积分} = \frac{1}{2^n \lambda} \cdot 2\pi i \cdot \left(\frac{n}{2} \right) = \frac{\pi}{2^{n-1}} \left(\frac{n}{2} \right)$$

$$(5) \int_0^{2\pi} \frac{dx}{2+\cos x} = \frac{1}{2} \int_0^{2\pi} \frac{dz}{1+\frac{1}{2}\cos z} = \frac{1}{2} \cdot \frac{2\pi}{\sqrt{1-\frac{1}{4}}} = \frac{2\pi}{\sqrt{3}}$$

$$(6) \int_0^\pi \frac{a dx}{a^2 + \sin^2 x} = \int_0^\pi \frac{a dx}{a^2 + \frac{1-\cos 2x}{2}} = \int_0^{2\pi} \frac{\frac{a}{2} dx}{a^2 + \frac{1-\cos x}{2}} = \int_0^{2\pi} \frac{a dx}{2a^2 + 1 - \cos x}$$

$$= \int_0^{2\pi} \frac{a dx}{2a^2 + 1 - \cos x} = \frac{1}{2a^2 + 1} \int_0^{2\pi} \frac{1}{1 - \frac{a}{2a^2 + 1} \cos x} = \frac{1}{2a^2 + 1} \cdot \frac{2\pi}{\sqrt{1 - \left(\frac{a}{2a^2 + 1}\right)^2}}$$

$$= \frac{2\pi}{\sqrt{(2a^2 - a + 1)(2a^2 + a + 1)}}$$

习题 1-21

$$(1) \int_{-\infty}^{+\infty} \frac{dx}{(x^2+a^2)^2(x^2+b^2)} = 2\pi i (\text{Res} f(ai) + \text{Res} f(bi)) = 2\pi i \left(\frac{-2ai}{(b^2-a^2)^2} + \frac{1}{(a^2-b^2)^2} \right)$$

$$\text{Res} f(ai) = \frac{1}{3!} \frac{d^3}{dz^3} \frac{1}{(z+ai)^2(z^2+b^2)} \Big|_{z=ai}$$

$$\text{Res} f(ai) = \frac{d^2}{dz^2} \frac{1}{(z+ai)^2(z^2+b^2)} \Big|_{z=ai} = \left[-2 \cdot \frac{1}{(z+ai)^3} - \frac{1}{z^2+b^2} - \frac{2z}{(z+ai)^2(z^2+b^2)^2} \right]_{z=ai}$$

$$= \frac{-2}{-8a^3 i (b^2-a^2)} - \frac{2ai}{(b^2-a^2)^2(-4a^2)} = \frac{1}{4a^2(b^2-a^2)} - \frac{1}{4a^3(b^2-a^2)} i \left(\frac{1}{2(b^2-a^2)^2 a} - \frac{1}{4a^3(b^2-a^2)} \right) i$$

$$\text{Res} f(bi) = \frac{1}{(a^2-b^2)^2(2bi)} = -\frac{1}{2(a^2-b^2)b} i$$

$$2\pi i (\text{Res} f(ai) + \text{Res} f(bi)) = 2\pi i \cdot \left(\frac{1}{2(a^2-b^2)^2 a} + \frac{1}{4a^3(a^2-b^2)} - \frac{1}{2(a^2-b^2)b} \right) i$$

$$= \left(\frac{1}{(a^2-b^2)^2 b} - \frac{1}{(a^2-b^2)^2 a} - \frac{1}{2a^3(a^2-b^2)} \right) \pi$$

$$(2) \int_{-\infty}^{+\infty} \frac{dx}{x(x-1)(x-2)} = \pi i \left(\frac{1}{3} - \frac{1}{2} + \frac{1}{1} \right) = 0$$

$$\begin{aligned}
 (3) \int_0^{\infty} \frac{x^{2n}}{x^{2n}+1} dx &= \frac{1}{2} \int_{-\infty}^{\infty} \frac{dx}{(x - e^{i(\frac{\pi}{2n})})(x + e^{i(\frac{\pi}{2n})})(x - e^{i(-\frac{\pi}{2n})})(x + e^{i(-\frac{\pi}{2n})})} \\
 &= \pi i (\text{Res} f(e^{i(\frac{\pi}{2n})}) + \text{Res} f(e^{i(\frac{\pi}{2n})})) = \pi i \left(\frac{1}{2e^{i(\frac{\pi}{2n})} \cdot i \cdot \sqrt{3}} + \frac{1}{2e^{i(\frac{\pi}{2n})} \cdot (-i) \cdot \sqrt{3}} \right) \\
 &= \frac{\pi}{2\sqrt{3}} (e^{i(-\frac{\pi}{2n})} - e^{i(-\frac{\pi}{2n})}) = \frac{\pi}{2}
 \end{aligned}$$

$$\begin{aligned}
 (4) \int_0^{\infty} \frac{x^{2m}}{x^{2n}+1} dx &= \frac{1}{2} \int_{-\infty}^{+\infty} \frac{x^{2m}}{x^{2n}+1} dx = \pi i \sum_{k=0}^{n-1} \text{Res} f(e^{i(\frac{\pi}{2n} + \frac{k\pi}{n})}) \\
 &= \pi i \sum_{k=0}^{n-1} \left. \frac{1}{2(n-m)x^{2(n-m)-1} - 2mx^{2m-1}} \right|_{e^{i(\frac{\pi}{2n} + \frac{k\pi}{n})}} \\
 &= \pi i \sum_{k=0}^{n-1} \left. \frac{x^{2m+1}}{2(n-m)x^{2n} - 2mx^{2m}} \right|_{e^{i(\frac{\pi}{2n} + \frac{k\pi}{n})}} \\
 &= \pi i \sum_{k=0}^{n-1} \frac{e^{i \cdot \frac{(2k+1)\pi(2m+1)}{2n}}}{-2n + 2m - 2m} = -\frac{\pi i}{2n} \sum_{k=0}^{n-1} e^{i \cdot \frac{(2k+1)(2m+1)\pi}{2n}} \\
 &= -\frac{\pi i}{2n} \cdot e^{i \cdot \frac{(2m+1)\pi}{2n}} \sum_{k=0}^{n-1} e^{i \cdot \frac{k(2m+1)\pi}{n}} \\
 &= -\frac{\pi i}{2n} e^{i \cdot \frac{(2m+1)\pi}{2n}} \cdot \frac{-1-1}{e^{i \cdot \frac{(2m+1)\pi}{n}} - 1} = \pi i \cdot e^{i \cdot \frac{(2m+1)\pi}{2n}} \cdot \frac{1}{e^{i \cdot \frac{(2m+1)\pi}{n}} - 1} \\
 &= \pi i \cdot \frac{1}{2i \cdot \sin[\frac{(2m+1)\pi}{n}]} = \frac{\pi}{2 \sin \frac{(2m+1)\pi}{n}}
 \end{aligned}$$

$$\begin{aligned}
 (5) \int_0^{\infty} \frac{x^2}{(x^2+a^2)^2} dx &= \frac{1}{2} \int_{-\infty}^{+\infty} \frac{x^2}{(x^2+a^2)^2} dx = \pi i \text{Res} f(ai) = \pi i \cdot \frac{d}{dz} \left. \frac{z^2}{(z+ai)^2} \right|_{ai} \\
 &= \pi i \cdot 2 \left[\left(\frac{z}{z+ai} \right) \cdot \frac{ai}{(z+ai)^2} \right]_{ai} = 2\pi i \cdot \frac{-a^2}{-8a^3 i} = \frac{\pi}{4a}
 \end{aligned}$$

習題 1.22

$$\begin{aligned}
 (1) \int_0^{\infty} \frac{1}{(x^2+a^2)(x^2+b^2)} \cos x dx &= \frac{1}{2} \int_{-\infty}^{+\infty} \frac{e^{ix}}{(x^2+a^2)(x^2+b^2)} dx = \pi i (\text{Res} f(ai) + \text{Res} f(bi)) \\
 &= \pi i \left(\frac{e^{-a}}{2ai(b^2-a^2)} + \frac{e^{-b}}{2bi(a^2-b^2)} \right) = \frac{\pi}{2} \left(\frac{e^{-a}}{a(b^2-a^2)} - \frac{e^{-b}}{b(b^2-a^2)} \right)
 \end{aligned}$$

$$\begin{aligned}
 (2) \int_{-\infty}^{\infty} \frac{e^{imx}}{x-i\alpha} dx &= \int_{-\infty}^{\infty} \frac{e^{imx}}{[x+i\text{Im}(\alpha)] - i\text{Re}(\alpha)} dx = e^{-im\text{Im}(\alpha)} \int_{-\infty}^{\infty} \frac{e^{imx}}{x-i\text{Re}(\alpha)} dx \\
 &= e^{-im\text{Im}(\alpha)} \left(\int_0^{\infty} \frac{e^{imx}}{x-i\text{Re}(\alpha)} dx + \int_0^{\infty} \frac{e^{-imx}}{-x-i\text{Re}(\alpha)} dx \right) = e^{-im\text{Im}(\alpha)}
 \end{aligned}$$

$$(2) \int_{-\infty}^{\infty} \frac{e^{imx}}{x - ia} dx = 2\pi i \cdot e^{-ma}$$

$$(3) \int_0^{\infty} \frac{\sin mx}{x(x^2+a^2)} dx \quad (m>0, a>0) = \frac{1}{2i} \int_{-\infty}^{+\infty} \frac{e^{imx}}{x(x^2+a^2)} dx = \cancel{\pi \operatorname{Res} f} \frac{1}{2i} (2\pi i \operatorname{Res} f(ia) + \pi i \operatorname{Res} f(0))$$

$$= \pi \operatorname{Res} f(ia) + \frac{\pi}{2} \operatorname{Res} f(0)$$

$$= \frac{\pi e^{-ma}}{-2a^2} + \frac{\pi}{2} \cdot \frac{1}{a^2} = \frac{\pi}{2a^2} (1 - e^{-ma})$$

$$(4) \int_0^{\infty} \frac{x \sin x}{1+x^2} dx = \pi \operatorname{Res} f(i) = \pi \cdot \frac{ie^{-1}}{(1+i)2i} = \frac{\pi}{2e}$$

$$(5) \int_0^{\infty} \frac{\cos mx}{1+x^4} dx = \pi i \left(\operatorname{Res} f\left(e^{i(\frac{\pi}{4})}\right) + \operatorname{Res} f\left(e^{i(\frac{3\pi}{4})}\right) \right)$$

$$= \pi i \cdot \left(\frac{e^{im \cdot e^{i(\frac{\pi}{4})}}}{4e^{i(\frac{\pi}{4})}} + \frac{e^{im \cdot e^{i(\frac{3\pi}{4})}}}{4e^{i(\frac{3\pi}{4})}} \right) = \frac{\pi i}{4} \cdot \left(\frac{e^{im \cdot (\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i)}}{e^{i(\frac{3\pi}{4})}} + \frac{e^{im \cdot (-\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i)}}{e^{i(\frac{\pi}{4})}} \right)$$

$$= \frac{\pi i}{4} \cdot e^{-\frac{\sqrt{2}}{2}m} \cdot \frac{e^{i(\frac{\sqrt{2}}{2}m + \frac{\pi}{4})} + e^{i(-\frac{\sqrt{2}}{2}m + \pi - \frac{\pi}{4})}}{1}$$

$$= \frac{\pi}{4} e^{-\frac{m}{\sqrt{2}}} \cdot \frac{e^{i(\frac{m}{\sqrt{2}} + \frac{\pi}{4})} - e^{i(-\frac{m}{\sqrt{2}} - \frac{\pi}{4})}}{i} = \frac{\pi}{2} e^{-\frac{m}{\sqrt{2}}} \sin\left(\frac{m}{\sqrt{2}} + \frac{\pi}{4}\right)$$

$$(6) \int_0^{\infty} \frac{\sin^2 x}{x^2} dx = \int_0^{\infty} \frac{1 - \cos 2x}{2x^2} dx = \frac{1}{2} \int_0^{\infty} \frac{1}{x^2} dx - \frac{1}{2} \int_0^{\infty} \frac{\cos 2x}{x^2} dx$$

$$(6) \int_0^{\infty} \frac{\sin^2 x}{x^2} dx = \int_0^{\infty} \frac{(e^{ix} - e^{-ix})^2}{2i} dx = -\frac{1}{4} \int_0^{\infty} \frac{e^{i \cdot 2x} + e^{-i \cdot 2x}}{x^2} dx$$

$$\int_0^{\infty} \frac{\sin^2 x}{x^2} dx = - \int_0^{\infty} \sin^2 x d\left(\frac{1}{x}\right) = - \left(\frac{\sin^2 x}{x} \Big|_0^{\infty} - \int_0^{\infty} \frac{1}{x} d(\sin^2 x) \right) = + \int_0^{\infty} \frac{\sin 2x}{x} dx$$

$$= \int_0^{\infty} \frac{\sin x}{x} dx = \frac{1}{2i} \int_{-\infty}^{+\infty} \frac{e^{ix}}{x} dx = \frac{\pi}{2} \operatorname{Res} f(0) = \frac{\pi}{2}$$